Writing and Interpreting Circle Equations

Reading the center and radius from an equation, writing an equation from a center and radius, completing the square, and applying circle equations

Reference for every problem on this sheet:

- Standard form: $(x-h)^2+(y-k)^2=r^2$ — center (h,k), radius r. The number on the right is r^2, never r.
- A point lies on the circle exactly when its coordinates make the left side equal r^2.
- General form $x^2+y^2+Dx+Ey+F=0$ becomes standard form by completing the square in x and in y.
- Give exact values unless the problem asks for a decimal, and state the unit named in the problem.

1. An air-traffic radar screen shows every plane within range of the tower. The edge of the range is the circle $(x-4)^2+(y+3)^2=49$, where x and y are in miles.

(a) Write the coordinates of the tower (the center of the circle).

Answer: ________________

(b) Find the radius of the radar range, in miles.

Answer: ________________ mi

(c) A plane is at $(11,-3)$. Show by computing a distance that it is exactly on the edge of the range.

Answer: ________________

2. A sprinkler stands at the point $(-2, 5)$ on a garden plan, with distances in meters. It waters every point within 6 meters of itself. Write the equation of the circle that forms the boundary of the watered region, in standard form.

Answer: ________________

3. On a map with distances in kilometres, a cell tower stands at $(3, -2)$. Its signal reaches exactly as far as a town at $(8, -2)$, and no farther.

(a) Find the radius of the coverage circle, in kilometres.

Answer: ________________ km

(b) Write the equation of the coverage boundary in standard form.

Answer: ________________

4. A stone dropped in a pond makes a ripple whose edge, at one instant, is $x^2+(y-6)^2 = 20$, with x and y in meters. Write the center of the ripple, then find its radius in meters, rounded to two decimal places.

Answer: ________________ m

5. A circular running track is described by

$$x^2 + y^2 - 6x + 8y - 11 = 0$$

with x and y in meters. Complete the square in x and in y to rewrite the equation in standard form, then state the center and the radius of the track in meters.

Answer: ________________ m

6. A circular fence is modelled by $(x-2)^2+(y-1)^2=25$, in meters, and a straight path runs along the line $y=4$. Find both x-coordinates where the path meets the fence. Show the equation you solved after substituting $y=4$.

Answer: ________________

7. Rewrite

$$x^2+y^2+10x-4y+20=0$$

in standard form by completing the square, then state the center and the radius in units.

Answer: ________________ units

8. A classmate looks at $x^2 + y^2 + 4x + 6y + 20 = 0$ and says "the center is $(-2, -3)$ and the radius is 20."

(a) Complete the square in x and in y and find the value that ends up on the right-hand side.

Answer: ________________

(b) Explain in two or three complete sentences what that value tells you about the graph of the equation, and say what is wrong with your classmate's claim.